FRIEDA: Benchmarking Multi-Step Cartographic Reasoning in Large Vision Language Models

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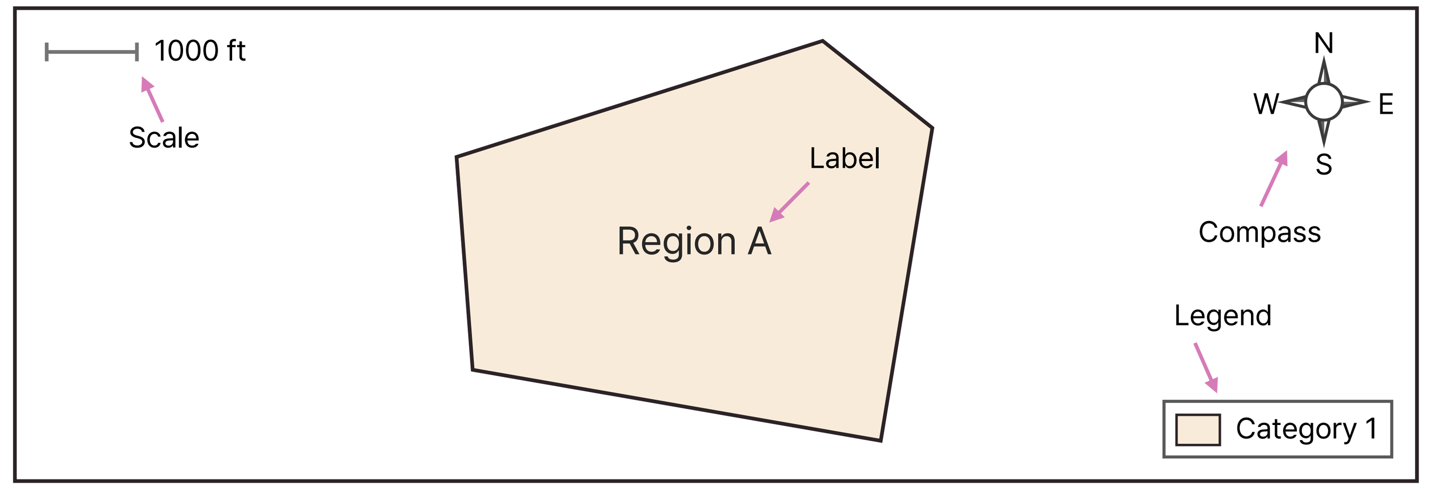
Webpage

What is FRIEDA?

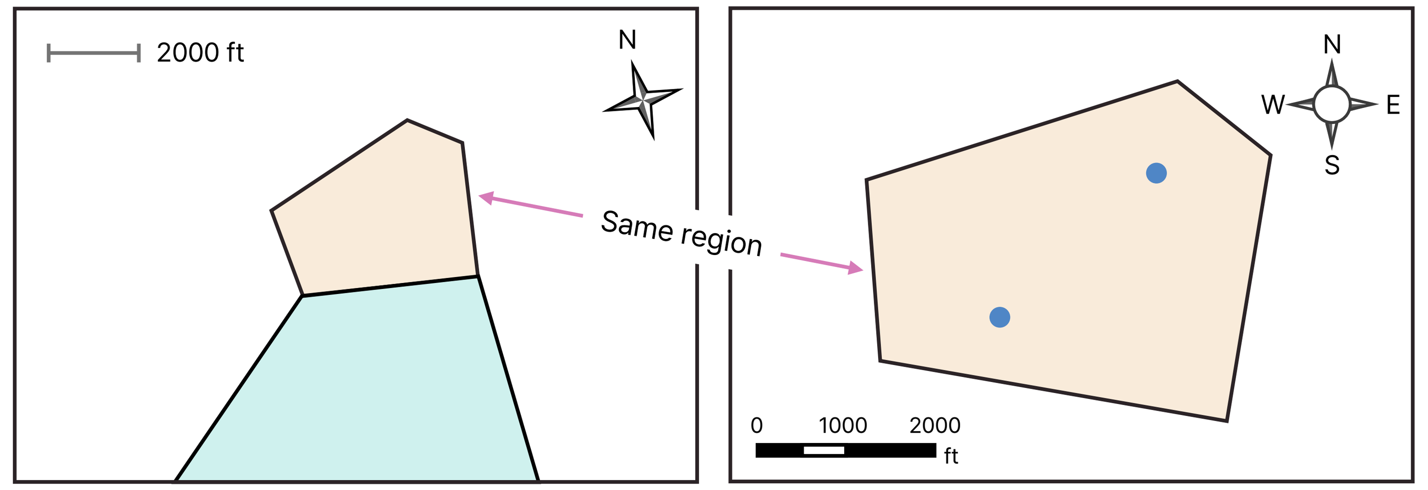
- Benchmark for **multi-step cartographic reasoning** on maps from real reports
- Reflects combination of **map-reading** and **general reasoning skills** people acquire over time
- Questions organized around a **transferable taxonomy of skills** curated by us, informed by:
  - GIS / cartography literature*, education*, map-user experience*

Cartographic Reasoning Skills

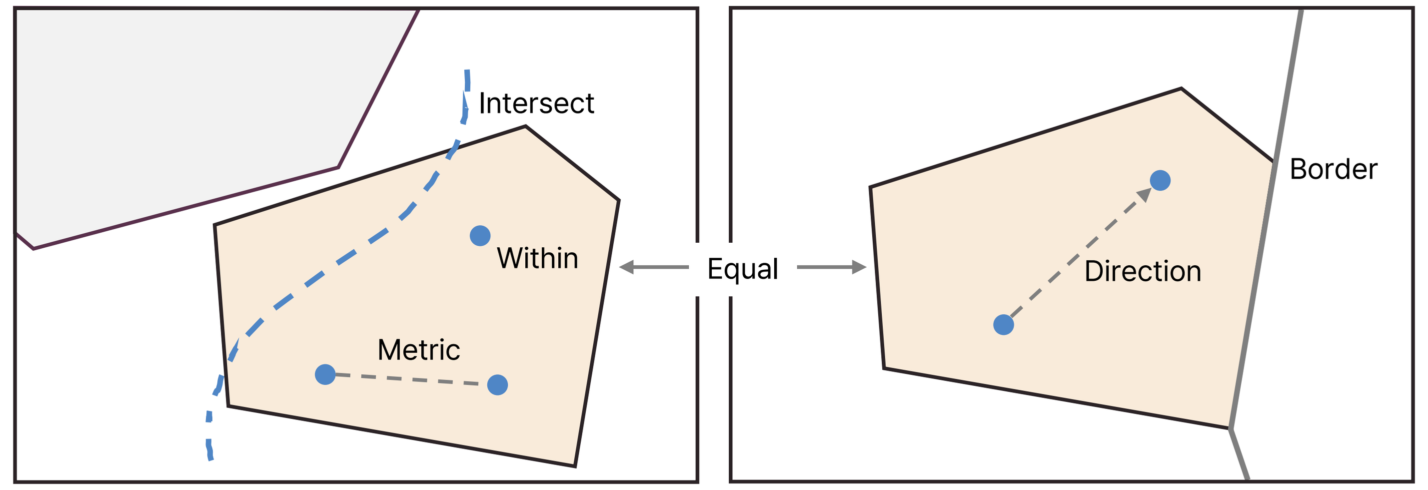
- Interpret map conventions*



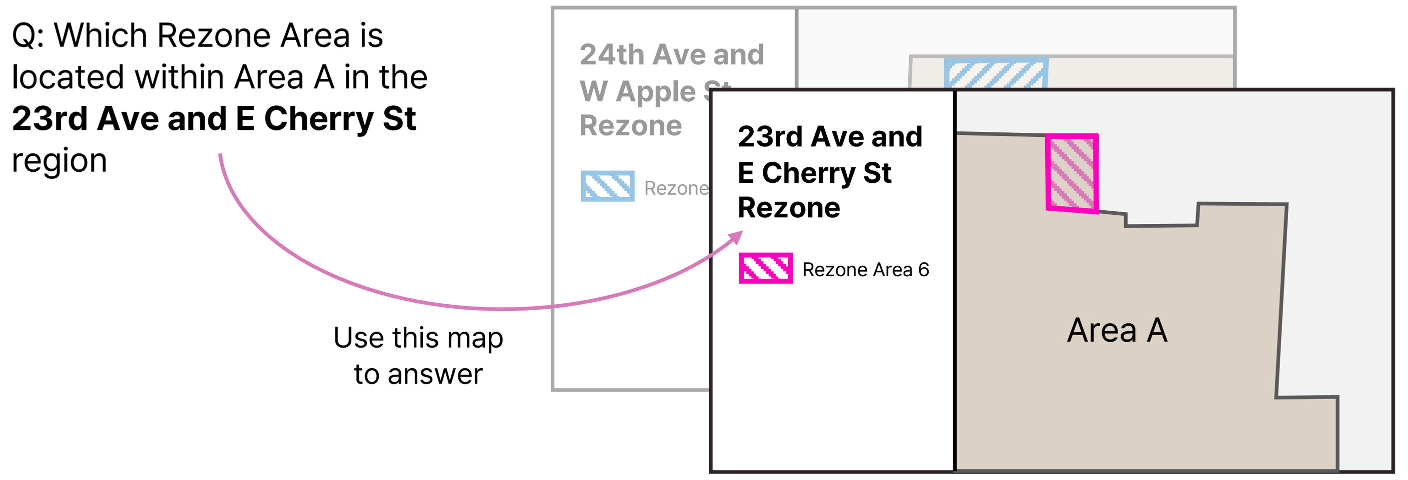
- Align evidence across maps**



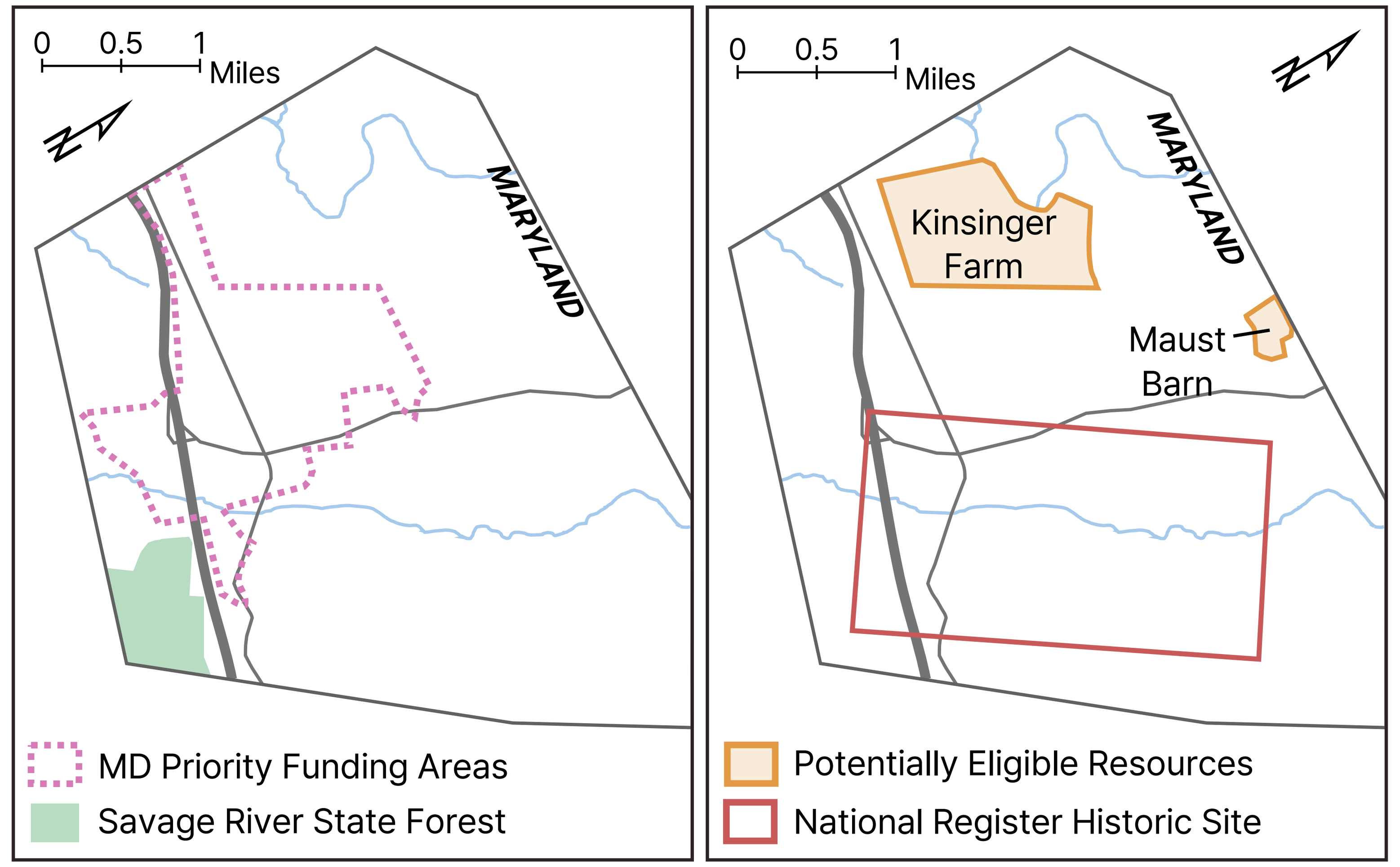
- Reason over spatial relations**



- Select correct maps*



Sample Question



What is the name of the Potentially Eligible Resources area that shares a border with the MD Priority Funding Area?

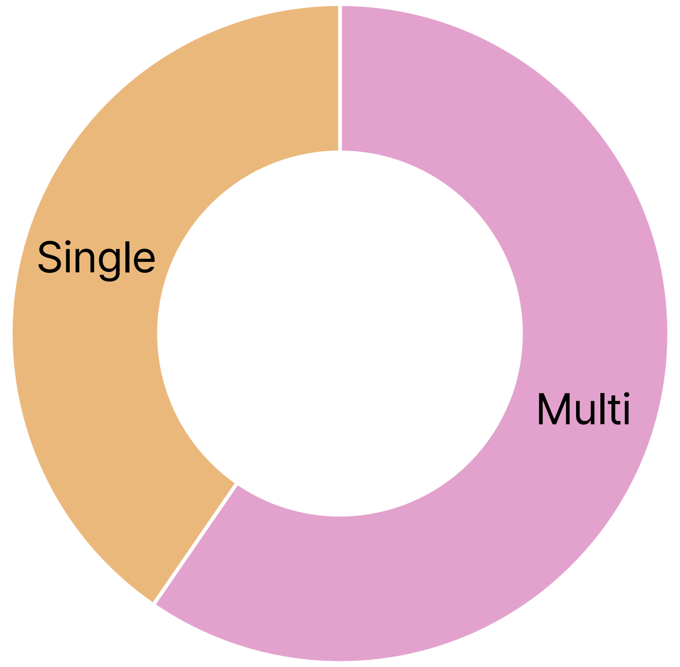
Steps:

1. Identify MD Priority Funding Area.
2. Identify all Potentially Eligible Resources: Kinsinger Farm, Maust Barn
3. Overlay the two maps
4. Identify the bordering region: Kinsinger Farm

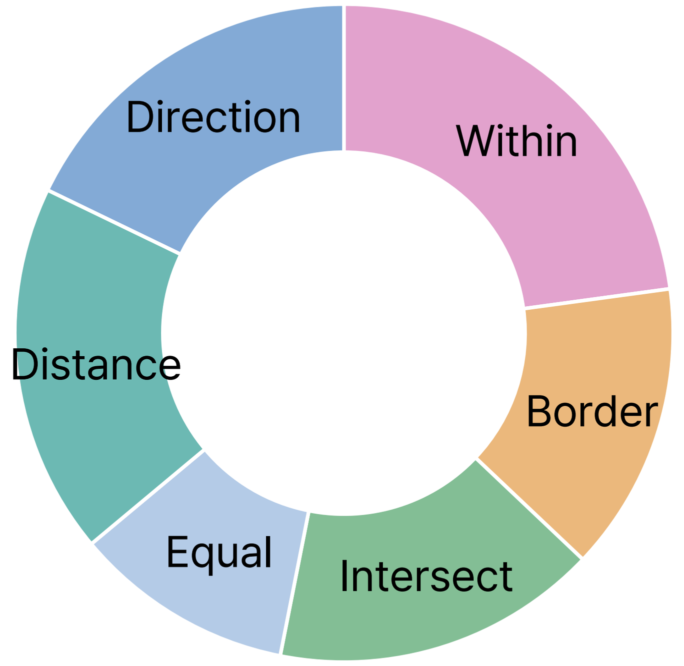
Answer: **Kinsinger Farm**

Dataset Statistics

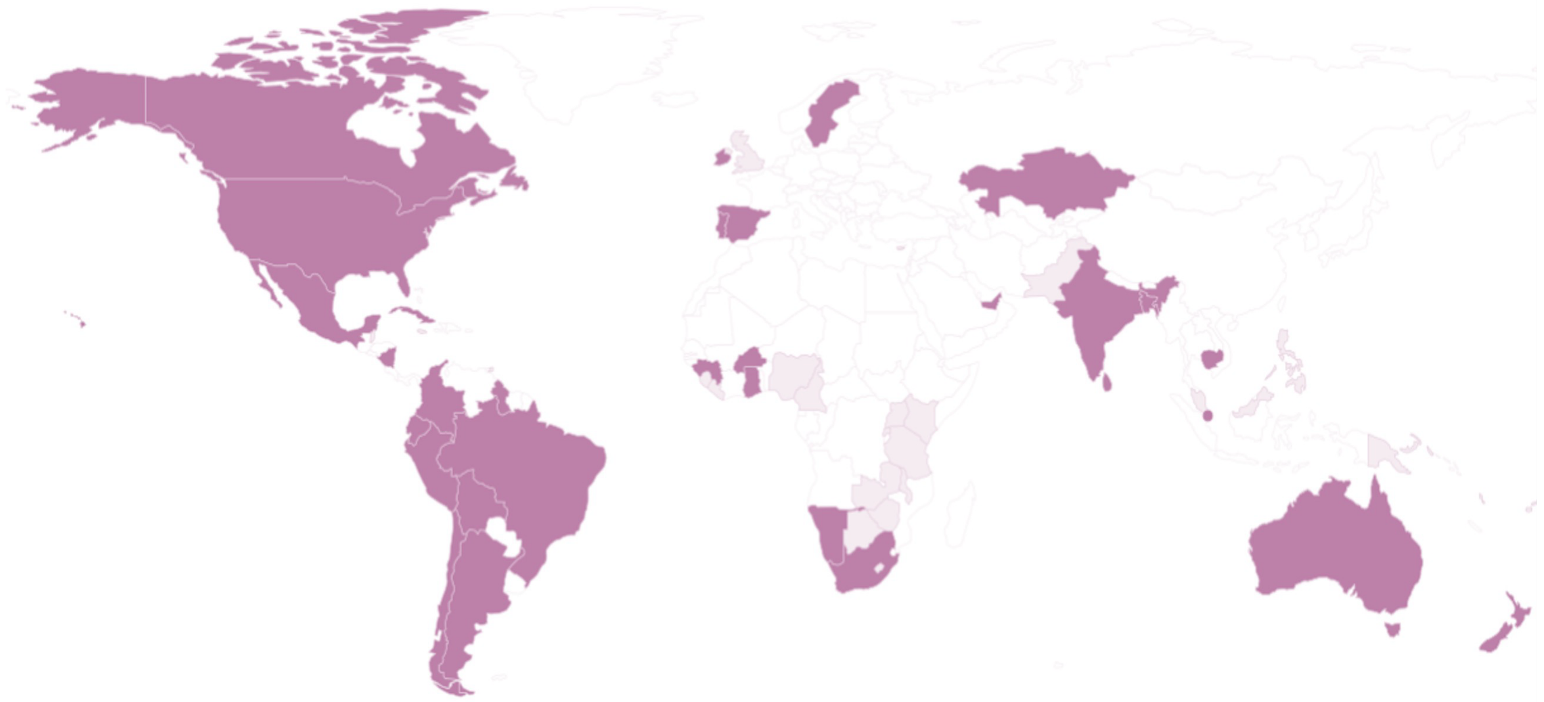
Map Count



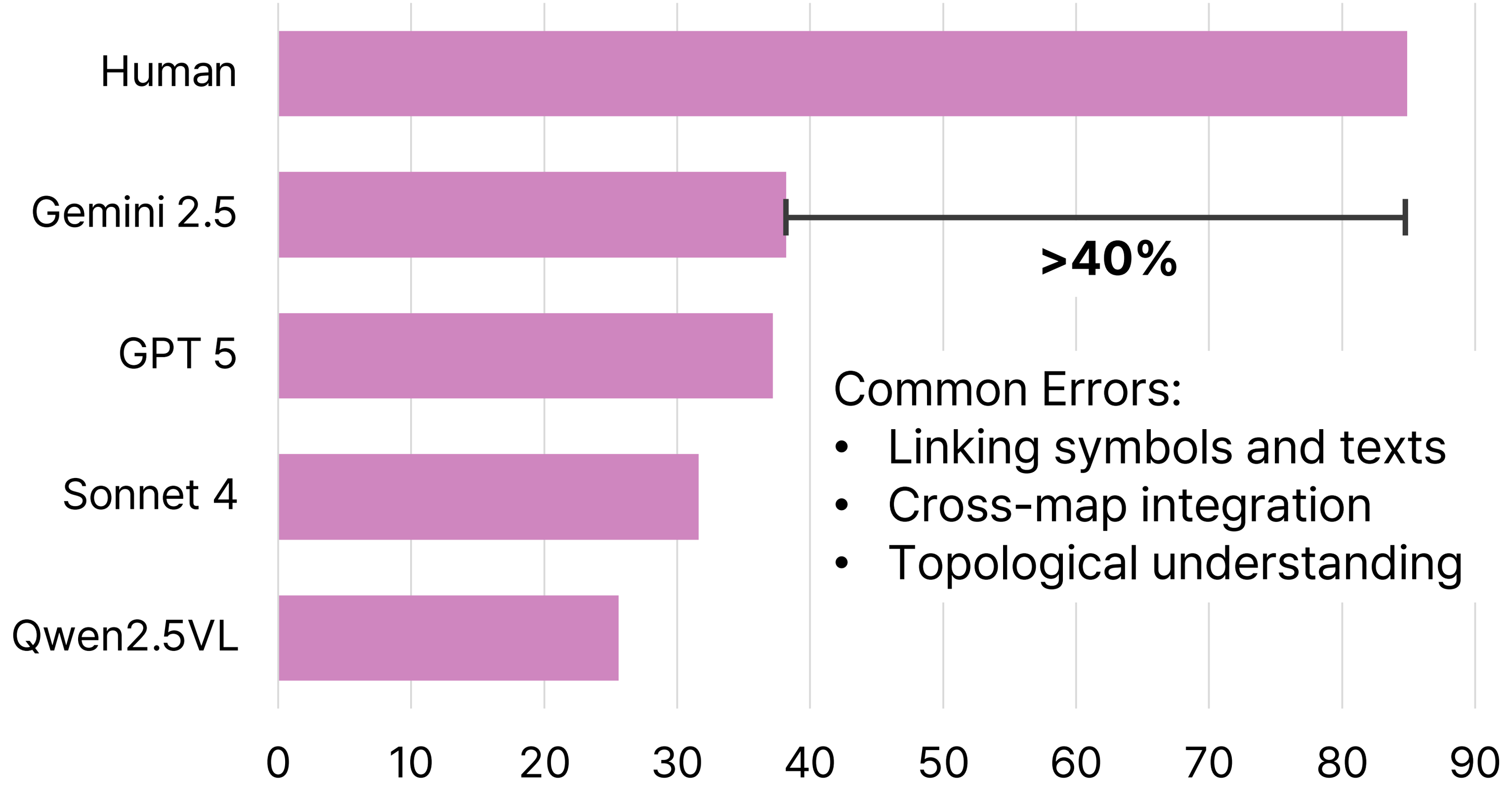
Spatial Relation



Country Distribution



Evaluation



Common Errors:

- Linking symbols and texts
- Cross-map integration
- Topological understanding

Curation Process